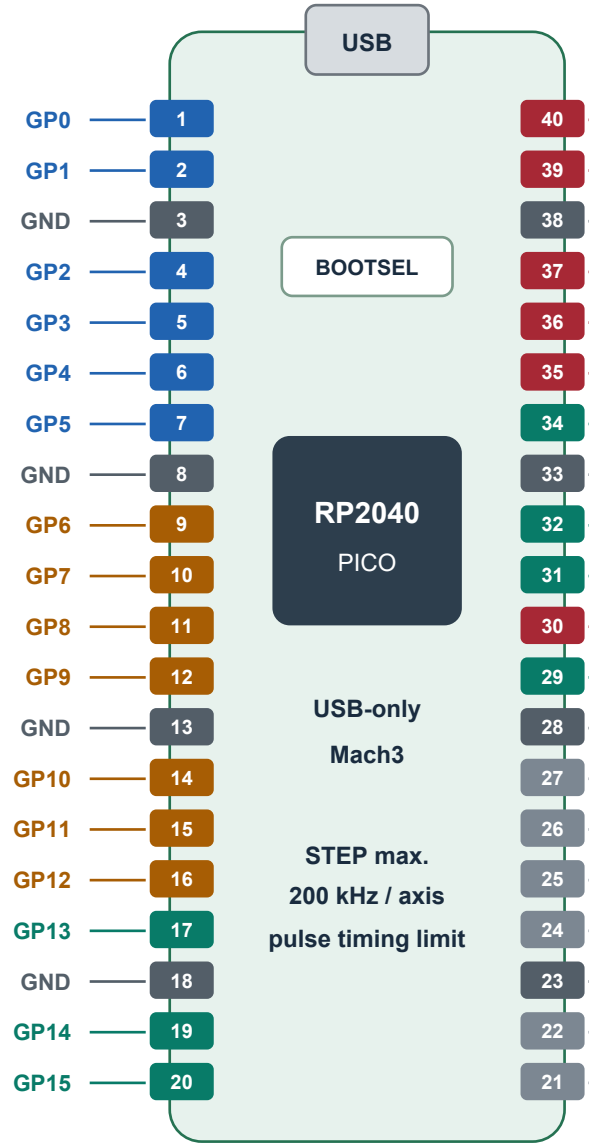
Pico pinout

Top view, USB connector at the top. Physical pin numbers are different from GPIO numbers.

Function	GPIO		GPIO	Function
X STEP	GP0	1	VBUS	USB 5 V
X DIR	GP1	2	VSYS	Supply input
Common ground	GND	3	GND	Common ground
Y STEP - mill only	GP2	4	3V3_EN	3.3 V enable - not GPIO
Y DIR - mill only	GP3	5	3V3_OUT	3.3 V supply output
Z STEP	GP4	6	ADC_VREF	ADC reference
Z DIR	GP5	7	GP28	DI7 - general IN
Common ground	GND	8	AGND	Common ground
PWM - spindle	GP6	9	GP27	DI6 - general IN
DO1 - motor ENABLE	GP7	10	GP26	DI5 - probe
DO2 - spindle RUN	GP8	11	RUN	Reset - not GPIO
DO3 - spindle DIR	GP9	12	GP22	DI4 - Z HOME + LIMIT
Common ground	GND	13	GND	Common ground
DO4 - mist coolant	GP10	14	GP21	Unassigned
DO5 - flood coolant	GP11	15	GP20	Unassigned
DO6 - general OUT	GP12	16	GP19	Unassigned
DI1 - E-stop	GP13	17	GP18	Unassigned
Common ground	GND	18	GND	Common ground
DI2 - X HOME + LIMIT	GP14	19	GP17	Unassigned
DI3 - Y HOME + LIMIT	GP15	20	GP16	Unassigned



Lathe: X = GP0/GP1; Z = GP4/GP5. The Y pair (GP2/GP3) is unused.

GPIO uses 3.3 V logic only. Use a suitable interface for 5 V / 24 V machine signals.

Wiring overview

Signal connections only. This diagram does not show motor power wiring.

PC / Windows

Mach3 - USB - COM port

USB →

Raspberry Pi Pico

RP2040 - 40 pin - USB-only

Pin numbers = physical pins

Motor-driver signal inputs

X axis

GP0 / pin 1 → **STEP IN**

GP1 / pin 2 → **DIR IN**

Mill and lathe

Y axis

GP2 / pin 4 → **STEP IN**

GP3 / pin 5 → **DIR IN**

Y: mill only

Z axis

GP4 / pin 6 → **STEP IN**

GP5 / pin 7 → **DIR IN**

Mill and lathe

Inputs

DI1 GP13/p17 - E-stop DI2 GP14/p19 - X home/limit

DI3 GP15/p20 - Y DI4 GP22/p29 - Z home/limit

DI5 GP26/p31 - Probe DI6 GP27/p32 - DI7 GP28/p34

Active low: GPIO - switch contact - GND



Outputs

ENABLE: GP7/p10 - PWM: GP6/p9

Spindle RUN: GP8/p11 - DIR: GP9/p12

Mist: GP10/p14 - Flood: GP11/p15 - DO6: GP12/p16

Never connect 5 V or 24 V directly to a Pico GPIO.

3.3 V signals - suitable interface - driver / VFD

Motor / VFD power is separate from Pico GPIO.

Mach3 DI1..DI7: port 1 / pins 1..7. These are logical input numbers, not physical Pico pin numbers.

Mach3 DI1..DI7 mapping: port 1 / pins 1..7. Inputs use pull-ups. Verify actual machine interfaces.